Homotopical Encryption: Hidden Hodge Operators and the Harmonic Cochain Problem

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Abstract

We introduce the Harmonic Cochain Problem (HCP): recovering the operative harmonic component $h = P_{\ker(\Delta_1)}b$ from public data (α, b). The degree-one Hodge–Laplacian Δ_1 is induced by an operative hidden representative $x \in A_\alpha = \pi^{-1}(\alpha)$ of the public classifier value α. Its adjacent coboundaries δ_0, δ_1 determine the Hodge decomposition $b = h + e$, with $e \in \mathrm{im}(\delta_0) \oplus \mathrm{im}(\delta_1^*)$. We compare HCP with the closest-vector problem (CVP), distinguishing the field-linear image $\mathcal{V}_B(R) = \{Bx : x \in R^n\}$ from the discrete lattice $\Lambda_B = B\mathbb{Z}^n$. The former can be realized as a linear kernel. For an explicitly supplied inclusion $\Lambda_B \subseteq W$, orthogonal projection onto the harmonic subspace W preserves the closest lattice vectors. We then prove a hidden-operator separation: if publicly equivalent hidden operators disagree on harmonicity, a correct known-lattice CVP oracle cannot supply a uniformly exact public-data HCP solver.

Contents

1 Introduction **3**
1.1 Motivation and Contributions 3

2 The Harmonic Cochain Problem **5**
2.1 Hodge Encryption and Correctness 9
2.2 Public and Hidden Data 11
2.2.1 Full projective observations 15
2.3 Recovery Target 16
2.4 Computational Scope 17

3 CVP Comparison and Kernel Ambiguity **17**
3.1 Known-Lattice CVP 17
3.2 Presented Linear Images 18

3.3 Explicit Harmonic-Lattice Bridge 20
3.4 Oracle Insufficiency 21
3.5 A Constructed Integral Cellular Family 22

4 Conclusion and Further Work **24**
4.1 Further Questions 25

A Statement–Lean Parity Table **25**

B Formal Objects and Proof Boundaries **31**

1 Introduction

1.1 Motivation and Contributions

Helfer's first-order homotopical logic [Hel25] provides the original semantic motivation for this framework: equality is interpreted through path objects and, under the stated hypotheses, closed-formula truth is invariant under homotopy equivalence of interpretations. This motivates distinguishing a hidden presentation from the semantic information retained by a public classifier. HCP asks how that distinction interacts with the harmonic target determined by a hidden Hodge operator.

Homotopical Encryption studies the recovery of an operative harmonic component when the Hodge operator is hidden. The public input specifies an observed cochain and a class of possible private presentations. In a known-lattice closest-vector problem, the lattice is presented before the search begins. In HCP, the recovery target is

$$h = P_H(b) \in \ker(\Delta_1),$$

but the public class does not include the degree-one Hodge–Laplacian Δ_1, its kernel, or the restriction maps from which Δ_1 is built. A solver must recover the harmonic cochain selected by the operative hidden Hodge operator.

The operator dependence can be summarized as follows. A hidden representative determines the restriction data, which determine the adjacent coboundaries and hence the degree-one Hodge operator:

$$x \longmapsto \{R^x_{\sigma\to\tau}\}_{\sigma\prec\tau} \longmapsto (\delta_{0,x}, \delta_{1,x}) \longmapsto \Delta_{1,x} \longmapsto \big(\ker \Delta_{1,x}, P_{H_x}\big).$$

The recovery operation is Hodge-theoretic. The public label records homotopy-class information about the hidden presentation. Homotopy-equivalent representatives share that label even when their Hodge operators differ in the fixed public coordinates. The public data can then leave the operative target undetermined.

The CVP comparison makes this dependence on the input precise. A real linear span can be realized as the kernel of a suitable linear operator, but a discrete lattice cannot equal a nonzero real kernel. If an explicit lattice Λ is supplied inside a harmonic subspace $W = \ker(\Delta_W)$, orthogonal projection of the target onto W preserves the closest vectors of Λ. This gives a solution-preserving comparison once the harmonic subspace and lattice inclusion are supplied. It does not recover the operative kernel from the public class.

Observation choice is part of the recovery problem. The setup in Section 2 distinguishes exact-component observations from full projective data of the operative kernel.

The principal notation is fixed globally as follows. Symbols used only inside a single proposition or construction are defined locally.

Symbol	Meaning
$\pi : A \to P$	homotopy-invariant public classifier
$\alpha = \pi(x) \in P$	realized public classifier value
$A_\alpha = [x]_\pi$	fiber of hidden representatives with public label α
(α, b)	public HCP observation
$a \in C^0,\ c = \delta_0 a \in C^1$	degree-zero opening and committed exact one-cochain
$\delta_0, \delta_1, \Delta_1$	adjacent coboundaries and degree-one Hodge–Laplacian
$\ker(\Delta_1),\ P_H$	harmonic subspace and harmonic projection
$\Delta_W = I - P_W$	kernel-normalized projection operator associated with W

Table 1: Core notation for the semantic interpretation and HCP.

This paper develops this boundary through the following statements.

(i) The Harmonic Cochain Problem (HCP) asks for recovery of the operative component $h = P_H b$ for a hidden degree-one Hodge operator Δ_1.

(ii) A Hodge-structured noise model gives the cochain decomposition $b = h + e$, where $h \in \ker(\Delta_1)$ and $e \in \operatorname{im}(\delta_0) \oplus \operatorname{im}(\delta_1^*)$.

(iii) Harmonic lattices in one public class can have different unique closest vectors to the same cochain, so known-lattice CVP access does not determine the operative answer.

(iv) Exact-component extraction can lose the full Hodge residual, whereas full projective data of the operative kernel determine its fixed-metric projector and harmonic target.

Finite Hodge theory follows the cellular-cochain framework of Hansen–Ghrist [HG19]; the algebraic model used here is specified in Section 2. Helfer's homotopy-invariant semantics [Hel25] motivates the public classifier; the constructed examples prove only the explicitly named nonzero-cohomology-test fragment.

2 The Harmonic Cochain Problem

Let A be the type of hidden presentations, with semantic interpretation M_x for each $x \in A$. Write $x \simeq y$ when M_x and M_y are homotopy equivalent in the declared semantics. Assume that this equivalence preserves the chosen closed tests. Let P be the type of public class data and choose a classifier $\pi : A \to P$ that factors through their closed theory. Then

$$x \simeq y \quad \Longrightarrow \quad \pi(x) = \pi(y).$$

The typed data and the invariance premise are made precise in Definition 2.12 and Proposition 2.13. For a realized public value α, write

$$\alpha \in \mathrm{im}(\pi), \qquad A_\alpha = \pi^{-1}(\alpha) = \{x \in A : \pi(x) = \alpha\}.$$

For $x \in A$, we also write $[x]_\pi := A_{\pi(x)}$. Thus, whenever $\alpha = \pi(x)$, the two descriptions agree: $A_\alpha = [x]_\pi$.

A classifier fiber may be coarser than a homotopy-equivalence class. A common public label specifies that fiber without supplying its operative representative or a transport map.

We attach Hodge data to each presentation using finite graded cell carriers, integer incidence numbers and face-to-coface linear stalk maps.

Fix a finite coordinate set N and let $V = \mathbb{R}^N$ have its Euclidean inner product. Each presentation has finite-dimensional real inner-product cochain spaces C^0_x, C^1_x, C^2_x. To compare presentations using one public cochain vector, fix isometric coordinate charts $\iota_x : C^1_x \xrightarrow{\sim} V$. We identify degree-one vectors and maps with their transported versions in V. The charts are comparison data; they are not asserted to be recoverable from the public classifier.

The restriction maps $R^x_{\sigma \to \tau}$ for $\sigma \prec \tau$, including vertex-to-edge and edge-to-face maps, determine adjacent coboundaries

$$\delta_{0,x} : C^0_x \to V, \qquad \delta_{1,x} : V \to C^2_x, \qquad \delta_{1,x}\delta_{0,x} = 0.$$

In these coordinates, define the degree-one Hodge–Laplacian, its harmonic subspace and its orthogonal projector by

$$\begin{aligned} \Delta_x &= \delta_{0,x}\delta^*_{0,x} + \delta^*_{1,x}\delta_{1,x}, \\ \mathcal{H}_x &= \ker(\Delta_x), \qquad P_{\mathcal{H}_x} : V \to \mathcal{H}_x \subseteq V. \end{aligned}$$

The stars denote adjoints. When the operative representative is fixed, we suppress x and also write Δ_1 or Δ for its degree-one Laplacian, and P_H for its harmonic projector.

In the Homotopical Encryption public-sample model, a presentation x produces a cochain

$$b = h + e, \qquad h \in \mathcal{H}_x, \qquad e \in \mathrm{im}(\delta_{0,x}) \oplus \mathrm{im}(\delta_{1,x}^*).$$

The message h is a harmonic representative of its degree-one cohomology class, and e is Hodge-structured noise. When x is also equipped with an explicit harmonic-lattice presentation, write $\Lambda_x \subseteq \mathcal{H}_x$ for its discrete lattice and choose $h \in \Lambda_x$. In this integral profile, the public fiber indexes a family of possible embedded harmonic lattices; the operative hidden member selects one of them. The real harmonic space and the supplied discrete lattice remain distinct objects.

Definition 2.1 (Public observation)**.** *A public HCP observation is a pair (α, b) consisting of a realized public classifier value $\alpha \in \mathrm{im}(\pi)$ and an observed degree-one cochain $b \in V$ in the fixed comparison coordinates.*

Definition 2.2 (Hidden Hodge witness)**.** *A hidden Hodge witness for a public observation (α, b) is a hidden representative $x \in A_\alpha$, satisfying $\pi(x) = \alpha$. Its presentation data select the operative operator Δ_x. The witness and b determine the coordinates*

$$h_x(b) := P_{\mathcal{H}_x} b, \qquad e_x(b) := b - h_x(b).$$

The Hodge decomposition in Proposition 2.7 gives

$$h_x(b) \in \ker(\Delta_x), \qquad e_x(b) \in \mathrm{im}(\delta_{0,x}) \oplus \mathrm{im}(\delta_{1,x}^*), \qquad b = h_x(b) + e_x(b).$$

Definition 2.3 (Harmonic Cochain Problem)**.** *The* Harmonic Cochain Problem *(HCP) asks, given the public observation (α, b), to recover the operative harmonic component*

$$h = h_x(b) \in \mathcal{H}_x,$$

equivalently the harmonic term in the orthogonal Hodge decomposition of b for the operative hidden witness $x \in A_\alpha$. The witness is bound to the instance but is not part of the public input.

The hidden-operator ambiguity studied below is the condition that the public data do not determine the operative Δ_x or its harmonic kernel. A concrete public artifact may commit to or certify values derived from this presentation without revealing the restriction maps or the induced operator.

The algebraic cellular model above adapts the finite Hodge setup of [HG19] without assuming a regular CW or simplicial realization for every

admitted incidence structure. All cellular examples below use this declared model; stronger topological realization and the separate simplicial-coherence conditions require additional data.

Definition 2.4 (Matrix harmonicity)**.** *For a hidden linear Hodge operator Δ, a cochain $h \in \mathbb{R}^N$ is Δ-harmonic when*

$$\Delta h = 0.$$

Thus the harmonic kernel is $\ker(\Delta)$*. In the projection model this agrees with the condition $h \in W$ for the kernel-normalized operator $\Delta_W = I - P_W$.*

Definition 2.5 (Kernel-normalized operator, harmonic projection, and residual)**.** *Let $V = \mathbb{R}^n$ be equipped with its standard inner product, and let $W \subset V$ be a linear subspace. Write*

$$P_W : V \to W \subset V \qquad \textit{and} \qquad P_{W^\perp} : V \to W^\perp \subset V$$

for the orthogonal projections. The kernel-normalized projection operator attached to W is

$$\Delta_W = I - P_W.$$

For $b \in V$, define the residual vector and residual score by

$$r_W(b) = \Delta_W b, \qquad \rho_W(b) = \|r_W(b)\|.$$

We say that b is W-harmonic when $r_W(b) = 0$. This operator records exactly the harmonic kernel and its orthogonal projection. It is not asserted to equal a cellular Hodge–Laplacian $\delta_0\delta_0^* + \delta_1^*\delta_1$, or to have the same positive spectrum, unless the cellular Laplacian additionally acts as the identity on $W^\perp$.

Proposition 2.6 (Projection harmonicity equivalences)**.** *For every $b \in V$,*

$$r_W(b) = P_{W^\perp} b, \qquad \rho_W(b) = \|P_{W^\perp} b\|, \qquad \rho_W(b) \geq 0.$$

Moreover

$$P_W b \in W, \qquad r_W(P_W b) = 0,$$

and the following conditions are equivalent:

$$r_W(b) = 0 \quad \Longleftrightarrow \quad \rho_W(b) = 0 \quad \Longleftrightarrow \quad b \in W.$$

Proof. Since $V = W \oplus W^\perp$ orthogonally, every $b \in V$ decomposes uniquely as

$$b = P_W b + P_{W^\perp} b.$$

Hence

$$(I - P_W)b = b - P_W b = P_{W^\perp} b,$$

giving $r_W(b) = P_{W^\perp} b$ and the displayed formula for $\rho_W(b)$. Nonnegativity is the nonnegativity of the norm. The projection $P_W b$ lies in W, so its orthogonal component is zero; therefore $r_W(P_W b) = 0$. Finally, $r_W(b) = 0$ iff $P_{W^\perp} b = 0$, which holds iff $b \in W$. The equivalence with $\rho_W(b) = 0$ follows from $\|x\| = 0$ iff $x = 0$. □

Proposition 2.7 (Hodge decomposition surface)**.** *Let the three consecutive cochain spaces be finite-dimensional real inner product spaces, and let the displayed stars denote adjoints. Suppose*

$$d_{\text{next}} d_{\text{prev}} = 0, \qquad \Delta = d_{\text{prev}} d_{\text{prev}}^* + d_{\text{next}}^* d_{\text{next}}.$$

Then every cochain b admits a decomposition

$$b = h + e_1 + e_2$$

with

$$\Delta h = 0, \qquad e_1 \in \operatorname{im}(d_{\text{prev}}), \qquad e_2 \in \operatorname{im}(d_{\text{next}}^*),$$

and

$$\langle h, e_1 \rangle = \langle h, e_2 \rangle = \langle e_1, e_2 \rangle = 0.$$

The decomposition is unique.

Proof. Each summand defining Δ is self-adjoint, so Δ is self-adjoint. In finite dimension, $\operatorname{im}(\Delta) = (\ker \Delta)^\perp$, and hence the ambient cochain space decomposes as

$$\ker(\Delta) \oplus \operatorname{im}(\Delta).$$

Since

$$\operatorname{im}(\Delta) \subseteq \operatorname{im}(d_{\text{prev}}) + \operatorname{im}(d_{\text{next}}^*),$$

every cochain b can be written as $h + e_1 + e_2$ with $h \in \ker \Delta$, $e_1 \in \operatorname{im}(d_{\text{prev}})$, and $e_2 \in \operatorname{im}(d_{\text{next}}^*)$. From $\Delta h = 0$, one obtains

$$\|d_{\text{prev}}^* h\|^2 + \|d_{\text{next}} h\|^2 = 0,$$

hence $d_{\text{prev}}^* h = 0$ and $d_{\text{next}} h = 0$. These identities give $\langle h, e_1 \rangle = 0$ and $\langle h, e_2 \rangle = 0$, while $d_{\text{next}} d_{\text{prev}} = 0$ gives $\langle e_1, e_2 \rangle = 0$. Hence the harmonic, exact, and coexact subspaces are pairwise orthogonal. Their sum is therefore direct, which makes the three components unique. □

2.1 Hodge Encryption and Correctness

For a hidden representative x, define its noise space by

$$\mathcal{E}_x = \mathcal{H}_x^{\perp}.$$

The Hodge decomposition identifies the noise space as the orthogonal direct sum

$$\mathcal{E}_x = \operatorname{im}(\delta_{0,x}) \oplus^{\perp} \operatorname{im}(\delta_{1,x}^*),$$

and gives the orthogonal decomposition

$$C_x^1 = \mathcal{H}_x \oplus^{\perp} \mathcal{E}_x.$$

Consequently,

$$\begin{aligned} \operatorname{combine}_x : \mathcal{H}_x \times \mathcal{E}_x &\longrightarrow C_x^1, & (h, e) &\longmapsto h + e, \\ \operatorname{split}_x : C_x^1 &\longrightarrow \mathcal{H}_x \times \mathcal{E}_x, & b &\longmapsto (P_{\mathcal{H}_x} b, P_{\mathcal{E}_x} b) \end{aligned}$$

are inverse linear bijections. Orthogonality also gives

$$\|h + e\|^2 = \|h\|^2 + \|e\|^2.$$

This bijection is the algebraic encryption/decryption construction. Given a harmonic message $h \in \mathcal{H}_x$, randomized Hodge encryption samples, or otherwise chooses, admissible noise $e \in \mathcal{E}_x$ and sets

$$\mathsf{Enc}_x(h; e) = \operatorname{combine}_x(h, e) = h + e.$$

Full coordinate recovery is split_x; message decryption retains its harmonic coordinate:

$$\mathsf{Dec}_x(b) = \operatorname{pr}_{\mathcal{H}_x}\big(\operatorname{split}_x(b)\big) = P_{\mathcal{H}_x} b.$$

For every admissible noise value,

$$\mathsf{Dec}_x\big(\mathsf{Enc}_x(h; e)\big) = h, \qquad \operatorname{split}_x\big(\operatorname{combine}_x(h, e)\big) = (h, e).$$

Thus the bijection is between the full message–randomness pair and C_x^1. After randomness is sampled, Enc_x is a randomized procedure and Dec_x is its message-correct left inverse; this does not assert that a randomized map from messages alone is a bijection.

The subscript x is essential. The hidden representative—and, in a keyed instantiation, the private material selecting or opening it—determines Δ_x, the two summands, and the splitting map. The correctness identity is pointwise and requires no uniform distribution on an infinite noise space. It is not an indistinguishability or one-wayness claim; those require separate assumptions about key generation, randomness, and the public view.

Exact-component observations. Fix one cellular Hodge system and an opening $a \in C^0$. Its exact one-cochain $c = \delta_0 a$ satisfies

$$P_H c = P_H \delta_0 a = 0.$$

Indeed, $\operatorname{im} \delta_0$ is orthogonal to the harmonic subspace. Thus this exact cochain is not the full sample $b = h + e$ when the harmonic message h is nonzero. An encoding modulo an integer, a digest of that encoding and a proof about its acceptance do not, by their types alone, supply the full b or its relation to the operative hidden operator.

There is also a distinction between the exact one-cochain and its opening. Linearity gives

$$\delta_0 a = \delta_0 a' \quad \Longleftrightarrow \quad a - a' \in \ker \delta_0.$$

Consequently the raw coboundary determines an opening only modulo $\ker \delta_0$. A unique degree-zero message encoding requires further application data. This elementary observation is independent of binding a specified exact C^1 payload cryptographically.

Proposition 2.8 (Exact-component obstruction)**.** *Suppose $v \in C^2$ and $\delta_1^* v \neq 0$. On the unrestricted Hodge-noise carrier, the samples*

$$b_t = t\delta_1^* v, \qquad t \in \mathbb{R},$$

have zero harmonic message and zero exact component, while

$$\|P_{\mathcal{H}^\perp} b_t\| = |t| \, \|\delta_1^* v\|.$$

Thus this full-observation residual does not factor through the exact component. If an extraction depends only on that component and its zero image has target distance zero, it cannot preserve any positive target far radius uniformly on this unrestricted carrier, at any source radius.

Proof. A coexact vector is orthogonal to $\mathcal{H}$, so its orthogonal projection onto $\mathcal{H}^\perp$ is itself. Homogeneity gives the displayed norm. The exact-component view is constantly zero, whereas this norm is zero at $t = 0$ and positive at $t = 1$, proving nonfactorization. For any fixed source radius, choose positive t large enough to exceed it. The extracted value still has distance zero, contradicting every positive target far radius. □

This obstruction concerns the unrestricted Hodge-noise carrier and an extraction depending only on its decoded exact component. Equal decoded components do not by themselves imply equal raw payload bytes for an

arbitrary decoder. A restricted noise or authenticated realization profile may exclude the family, so a comparison for such a profile requires its explicit hypotheses. The scores

$$\|P_{H^\perp} b\|, \qquad \operatorname{dist}(b, \operatorname{im} \delta_0), \qquad \text{a modular cochain or code distance}$$

ask different questions. The first measures disagreement with the harmonic subspace, the second distance to exactness, while a modular distance needs its own carrier and normalization. No equality of these scores is assumed by the HCP/CVP comparison.

2.2 Public and Hidden Data

The pair (α, b) is the vector-level public input. An accepted exact payload, an opening of its exact component or a proof about that component does not by itself expose the full cochain b. A record used to present this input must be bound to the operative operator, classifier and sample, with an explicit disclosure or specified extraction of the full cochain.

Recovery claims are relative to that chosen observation. If a larger released view $\mathsf{View}(x, b)$ is used, an ambiguity argument for two candidate presentations also requires equality of that view. A view factoring through (α, b) preserves equality of the public pair; arbitrary snapshot roots, geometric coordinates or proof transcripts need not. The full projective observation considered below gives a concrete example of such additional information.

Public equality is equality after applying the classifier, rather than identity of hidden presentations. The semantic equivalence under consideration must preserve this public label. The converse need not hold: equal labels do not supply an equivalence or its transport. Even when a fiber contains distinct representatives, that collision condition alone gives no one-wayness or indistinguishability guarantee.

Definition 2.9 (Hidden-operator public projection)**.** *For a public class type P, a public projection is a map*

$$\pi_\Delta : \{\textit{hidden Hodge operators } \Delta\} \longrightarrow P.$$

Two hidden operators are publicly equivalent when $\pi_\Delta(\Delta) = \pi_\Delta(\Delta')$. The hidden operator fiber over $\alpha \in P$ is

$$\operatorname{Fib}_{\pi_\Delta}(\alpha) = \pi_\Delta^{-1}(\alpha) = \{\Delta : \pi_\Delta(\Delta) = \alpha\},$$

and the hidden class of Δ is $\operatorname{Fib}_{\pi_\Delta}(\pi_\Delta(\Delta))$.

Definition 2.10 (Lossiness and kernel ambiguity)**.** *The public projection* π_Δ *is* lossy *if there exist* $\Delta \neq \Delta'$ *with* $\pi_\Delta(\Delta) = \pi_\Delta(\Delta')$*. The two operators have the same harmonic kernel if*

$$\forall v, \quad \Delta v = 0 \Longleftrightarrow \Delta' v = 0.$$

The projection is kernel-ambiguous *if there are publicly equivalent operators* Δ, Δ' *whose harmonic kernels are not the same.*

Definition 2.11 (Public-compatible Hodge ambiguity)**.** *Public-compatible Hodge ambiguity means that there exist publicly equivalent operators* Δ, Δ' *and a cochain-facing observation* b *such that*

$$\Delta b = 0 \qquad \textit{but} \qquad \Delta' b \neq 0.$$

The semantic input is a typed interpretation of a multi-sorted algebraic signature. Its sorts are interpreted by objects of a suitable model category, finite contexts by products, and function symbols by the corresponding maps. A homotopy equivalence of interpretations must respect those operations up to the specified homotopies and supply componentwise homotopy-equivalence data between cofibrant-fibrant interpretations.

Helfer's first-order semantics interprets equality through path objects. Under its stated categorical hypotheses, homotopy-equivalent interpretations satisfy the same closed formulas [Hel25, Definition 7.1 and Theorem 12.3]. We retain invariance of the chosen closed tests as an explicit premise of the generic semantic construction below. The full first-order theorem is external background; the cellular example in Proposition 3.5 establishes the corresponding premise for its declared homological tests.

Definition 2.12 (Typed HCP semantic realization)**.** *Fix a multi-sorted algebraic signature* σ*, a model category* $\mathcal{C}$ *with terminal object and binary products, and, for each hidden representative* $x \in A$*, a base interpretation* $M_x : \sigma \to \mathcal{C}_{\mathrm{f}}$ *together with homotopical formula semantics over* M_x*. Write*

$$\mathrm{Th}(M_x) = \{\varphi \textit{ closed} : M_x \vDash \varphi\}$$

for its closed first-order theory. Relative to the presentation data and coordinate charts fixed above, a typed HCP semantic realization consists additionally of maps

$$\begin{aligned} x &\longmapsto \Delta_x \in \mathrm{End}(V), \\ \kappa &: \{\textit{closed theories}\} \longrightarrow P, \\ \pi_\Delta &: \{\textit{Hodge operators}\} \longrightarrow P, \end{aligned}$$

satisfying the two factorization identities

$$\pi(x) = \kappa(\mathrm{Th}(M_x)), \qquad \pi_\Delta(\Delta_x) = \pi(x).$$

The Laplacian Δ_x is presentation-level Hodge data extending the semantic core M_x; it is not silently included among the symbols whose closed theory is asserted to be homotopy invariant.

Proposition 2.13 (Helfer-to-HCP classifier bridge)**.** *Let $x, y \in A$ belong to a typed HCP semantic realization. If M_x and M_y are homotopy equivalent interpretations in the sense of Helfer, then*

$$\pi(x) = \pi(y) \qquad \textit{and} \qquad \pi_\Delta(\Delta_x) = \pi_\Delta(\Delta_y).$$

If, independently, there is a cochain b for which

$$\Delta_x b = 0, \qquad \Delta_y b \neq 0,$$

then π_Δ has public-compatible Hodge ambiguity, and hence is kernel-ambiguous and lossy.

Proof. The stated closed-test invariance premise gives, for every closed σ-formula φ,

$$M_x \vDash \varphi \quad \Longleftrightarrow \quad M_y \vDash \varphi.$$

In the full first-order setting this premise is supplied by Helfer's homotopy-invariance theorem [Hel25, Theorem 12.3]. Hence $\mathrm{Th}(M_x) = \mathrm{Th}(M_y)$. Applying κ and then the two factorization identities gives

$$\pi(x) = \pi(y), \qquad \pi_\Delta(\Delta_x) = \pi_\Delta(\Delta_y).$$

The additional displayed harmonicity and non-harmonicity conditions now exhibit (Δ_x, Δ_y, b) as a public-compatible Hodge-ambiguity witness. The ambiguity hierarchy below supplies kernel ambiguity and lossiness. □

The direction in Proposition 2.13 is one-way. Homotopy-equivalent interpretations have the same factored public value, but equality of public values need not reconstruct a homotopy equivalence: a classifier fiber may be strictly coarser than an equivalence class. Likewise, the equivalence itself does not produce kernel disagreement. If Δ-harmonicity were included in the invariant first-order signature, and the common coordinate b were named compatibly in both interpretations, Helfer's theorem would instead rule out the displayed disagreement.

There is a second, geometric distinction. Suppose an equivalence of hidden presentations induces degreewise linear isometric equivalences U_0, U_1, U_2 satisfying

$$U_1\delta_{0,x} = \delta_{0,y}U_0, \qquad U_2\delta_{1,x} = \delta_{1,y}U_1.$$

After conjugating by the fixed coordinate charts, U_1 is an isometry of the common comparison space V. Because the U_k preserve inner products, the adjoints transport as well, and hence

$$\Delta_y U_1 = U_1\Delta_x, \qquad U_1(\ker \Delta_x) = \ker \Delta_y.$$

The corresponding orthogonal projections commute with transport:

$$U_1 P_{\ker \Delta_x} b = P_{\ker \Delta_y} U_1 b.$$

Thus a transported observation preserves both the operative target and harmonicity:

$$\Delta_x b = 0 \quad \Longleftrightarrow \quad \Delta_y(U_1 b) = 0.$$

The ambiguity witness in Proposition 2.13 deliberately compares the same untransported public coordinate $b \in V$, not the pair $(b, U_1 b)$. Under the displayed conjugacy, its extra separation condition is equivalently

$$b \in \ker \Delta_x, \qquad b \notin U_1(\ker \Delta_x).$$

It therefore requires a nonstabilization witness in addition to the equivalence, and it models a public view that neither reveals nor applies the private transport U_1. If the observation were transported together with the hidden presentation, this particular ambiguity would disappear.

Proposition 2.14 (Ambiguity hierarchy)**.** *Public-compatible Hodge ambiguity implies kernel ambiguity, and kernel ambiguity implies that the public projection is lossy.*

Proof. If b is harmonic for Δ but not for Δ', then the two operators cannot have the same harmonic kernel. This gives kernel ambiguity. If publicly equivalent operators have different kernels, they cannot be the same operator, since equality of operators implies equality of kernels. Therefore the public projection identifies two distinct hidden operators. □

The hidden data include the fibration representative, restriction maps, the induced Laplacian, its harmonic kernel, and private witness/cochain data. Public data may bind to this hidden data through commitments or proof-carrying artifacts, but it does not present the operative kernel.

Definition 2.15 (HCP solution)**.** *Relative to an operative hidden witness x, a solution to an HCP instance (α, b) is the cochain $h' = h_x(b) = P_{\ker(\Delta_x)}b$. A public-data-only solver is* uniformly exact *if the same output has this value for every hidden witness compatible with its public input. The hidden-operator separation concerns this uniform contract; it does not deny that decryption succeeds once the operative private geometry is supplied.*

2.2.1 Full projective observations

At the fixed ambient inner product, the full projective point of the operative harmonic subspace determines its projector and hence the target from b. A digest-only value is a different observation and is not covered by this reconstruction.

Let $V = \mathbb{R}^n$ have its fixed Euclidean inner product, and let $w_1, \ldots, w_r$ be an independent frame spanning $W \leq V$. Its exterior coordinate is

$$c = w_1 \wedge \cdots \wedge w_r \in \bigwedge^r V.$$

An independent frame has nonzero exterior product. A different basis of the same subspace changes c by the nonzero determinant of its change-of-basis matrix. The projective class therefore depends only on W:

$$c \sim c' \quad \Longleftrightarrow \quad \exists u \in \mathbb{R}^\times,\ c' = uc.$$

The formal raw carrier is the scalar-orbit quotient of the whole exterior power, including its zero orbit. A genuine frame determines a nonzero, decomposable orbit. The conclusions below use such frames; they do not assert that every point of the raw quotient is a Grassmannian point.

A related specialized reconstruction uses restricted Plücker coordinates for maximal isotropic subspaces [DK26]. The argument here uses the full projective point of an arbitrary real subspace and the fixed ambient metric; no orthogonal-matroid realization is assumed.

Proposition 2.16 (Projective Plücker invariance)**.** *Two independent frames spanning the same finite-dimensional subspace have equal projective exterior coordinates.*

Proof. Both exterior products are nonzero elements of the one-dimensional top exterior power of that subspace. They differ by a nonzero scalar and hence have the same orbit. □

Recovery from the full point.

Proposition 2.17 (Recovery from the full projective point)**.** *Let $w_1, \dots, w_r$ be an independent frame for $W \subseteq \mathbb{R}^n$, and put $c = w_1 \wedge \cdots \wedge w_r$. Then*

$$W = \{x \in \mathbb{R}^n : x \wedge c = 0\}.$$

This reconstruction depends only on $[c]$. Consequently, two genuine rank-r frame witnesses have the same full projective point if and only if they present the same subspace. At the fixed ambient inner product, the full point determines P_W and $I - P_W$.

Proof. If $x \in W$, the frame with x adjoined is dependent and its exterior product vanishes. If $x \notin W$, the adjoined frame is independent and its exterior product is nonzero. Multiplying c by a nonzero scalar does not change this annihilator. The projector is the unique orthogonal projection onto the reconstructed subspace. □

In the raw quotient the zero orbit has the whole ambient space as its annihilator. The frame-indexed statement avoids treating that orbit as an arbitrary nonzero rank-r Plücker point. If $W = \ker \Delta_x$ is the operative harmonic subspace, full projective data together with b determine

$$h = P_W b.$$

The operative-kernel identification and the fixed inner product are essential. A coordinate for an unrelated subspace gives no such conclusion. The full point does not determine the positive spectrum of the original Δ_x. Nor does this exact-real reconstruction give an efficient finite byte algorithm or inversion of a digest.

2.3 Recovery Target

The recovery target is the orthogonal projection of b into the operative kernel, not the entire kernel subspace. Thus CVP asks for a closest vector in a presented lattice, while HCP asks for $P_{\ker(\Delta_x)} b$ for the operative hidden representative.

In the metric variant used for comparison with CVP, the continuous target is the closest harmonic cochain to b. Presenting Δ or a basis for $\ker(\Delta)$ presents a real subspace, not a discrete lattice. A genuine CVP instance additionally requires an explicit lattice $\Lambda \subseteq \ker(\Delta)$; Proposition 3.2 shows that projection onto the kernel preserves the closest vectors of that supplied lattice. The hidden-operator model withholds the operative subspace itself.

2.4 Computational Scope

The comparisons below concern exact recovery and preservation of discrete CVP solutions when the harmonic subspace, lattice and inclusion are supplied. The ambiguity examples show that the stated public input can fail to identify its operative target, ruling out a uniformly exact solver on that family. A computational hardness assumption requires encoded instance families, a specified measure of input size, an algorithm class and a success criterion; the present results do not constitute an asymptotic CVP-hardness reduction. Distributional encryption security would further require a sampling model and a defined adversary view.

3 CVP Comparison and Kernel Ambiguity

CVP asks for a closest vector in an explicitly presented lattice [MG02]. HCP asks for the projection of b into the kernel of the operative hidden Hodge operator. In the Homotopical Encryption model, public data need not determine the corresponding harmonic kernel.

3.1 Known-Lattice CVP

An explicit Euclidean CVP instance consists of a full-column-rank generator matrix $B \in \mathbb{R}^{m\times n}$ and a target $t \in \mathbb{R}^m$. Its discrete lattice is

$$\Lambda_B = B\mathbb{Z}^n = \{Bz : z \in \mathbb{Z}^n\}.$$

The injectivity of $z \mapsto Bz$ records that the displayed columns are a lattice basis. A correct closest-vector answer is a vector $v \in \Lambda_B$ such that

$$\|t - v\|^2 \le \|t - w\|^2 \qquad \text{for every } w \in \Lambda_B.$$

Thus, for fixed dimensions, a CVP oracle O is correct when

$$\begin{aligned} \mathsf{CVP}_{\mathrm{corr}}(O) \quad \Longleftrightarrow \quad & \forall \text{ full-column-rank } B, t, \\ & O(B,t) \in \Lambda_B \quad \text{and} \quad \forall w \in \Lambda_B, \\ & \|t - O(B,t)\|^2 \le \|t - w\|^2. \end{aligned}$$

The weaker output condition is merely lattice membership:

$$\forall B, t,\ O(B,t) \in \Lambda_B.$$

Fix a class $\mathcal{C}$ of admissible algorithms, such as polynomial-time or probabilistic polynomial-time algorithms. The external CVP hardness assumption is the schema

$$\mathsf{CVP}_{\mathrm{hard}}(\mathcal{C}) \quad \Longleftrightarrow \quad \neg\exists O,\ O \in \mathcal{C} \wedge \mathsf{CVP}_{\mathrm{corr}}(O).$$

A CVP oracle receives B and t. The HCP procedures studied here receive (α, b); the operative Hodge operator and its kernel are withheld.

3.2 Presented Linear Images

Let R be a field. The field-linear image of B is a different object:

$$\mathcal{V}_B(R) = BR^n = \{Bx : x \in R^n\}.$$

Over a field this is a linear subspace, not the discrete lattice $B\mathbb{Z}^n$. The following comparison is therefore linear algebra; it is not a reduction from discrete CVP.

Let B present $\mathcal{V}_B(R)$. If a matrix Δ satisfies $\ker(\Delta) = \mathcal{V}_B(R)$, then any solver satisfying

$$\Delta S(\Delta, t) = 0$$

returns $S(\Delta, t) \in \mathcal{V}_B(R)$.

Proposition 3.1 (Kernel realization for explicit generated subspaces)**.** *Let $B \in R^{n\times n}$ be an explicit square generator matrix over a field and set $\mathcal{V}_B(R) = \{Bx : x \in R^n\}$. There exists a matrix Δ such that*

$$\ker(\Delta) = \mathcal{V}_B(R).$$

Consequently, any solver $S(\Delta, b)$ whose output is Δ-harmonic for every presented Δ returns a vector in the explicit generated subspace $\mathcal{V}_B(R)$ when run on this realized operator.

Proof. Let $B \in R^{n\times n}$ be an explicit generator matrix over a field R, set

$$\mathcal{V}_B(R) = \{Bx : x \in R^n\} \subseteq R^n,$$

and let $t \in R^n$ be the solver input. We construct a matrix Δ with

$$\ker(\Delta) = \mathcal{V}_B(R).$$

Put $V = \mathcal{V}_B(R)$. Since R^n is finite-dimensional over a field, choose a complementary subspace W with

$$V + W = R^n, \qquad V \cap W = 0.$$

Thus every $y \in R^n$ has a unique decomposition

$$y = v + w, \qquad v \in V, \quad w \in W.$$

Equivalently, there is a unique element $w_y \in W$ such that $y - w_y \in V$. Define

$$P(y) = w_y.$$

The uniqueness of the $V \oplus W$ decomposition gives linearity: for $y, z \in R^n$, the vector $P(y) + P(z)$ lies in W, and

$$(y + z) - (P(y) + P(z)) = (y - P(y)) + (z - P(z)) \in V,$$

so $P(y + z) = P(y) + P(z)$. Similarly, for $a \in R$, $aP(y) \in W$ and

$$ay - aP(y) = a(y - P(y)) \in V,$$

so $P(ay) = aP(y)$.

We now compute the kernel. If $y \in V$, then $0 \in W$ and $y - 0 \in V$, hence uniqueness gives $P(y) = 0$. Conversely, if $P(y) = 0$, then $y - P(y) = y \in V$. Therefore

$$\ker P = V = \mathcal{V}_B(R).$$

Let Δ be the matrix of the linear map P. Then, for every $y \in R^n$,

$$\Delta y = 0 \quad \iff \quad y \in \ker P \quad \iff \quad y \in \mathcal{V}_B(R) \quad \iff \quad \exists x,\ Bx = y.$$

Let $S : R^{n \times n} \times R^n \to R^n$ be a presented-operator Hodge solver satisfying

$$\forall \Delta\, b, \qquad \Delta(S(\Delta, b)) = 0.$$

Applying this to the Δ constructed above and $b = t$ gives

$$\Delta(S(\Delta, t)) = 0.$$

By the kernel calculation, there exists x with

$$Bx = S(\Delta, t).$$

Hence $S(\Delta, t) \in \mathcal{V}_B(R)$. □

When $R = \mathbb{R}$, if a presented-operator solver additionally minimizes Euclidean distance among Δ-harmonic vectors, the same kernel equality says that it returns the continuous closest point of $\mathcal{V}_B(\mathbb{R})$. This is still not discrete CVP unless a separate lattice is supplied.

3.3 Explicit Harmonic-Lattice Bridge

The discrete bridge requires both kinds of data: a real harmonic subspace and an integer lattice contained in it. Let $W \subseteq \mathbb{R}^m$ be the operative harmonic subspace, let P_W be its orthogonal projection, and let $\Lambda_B = B\mathbb{Z}^n \subseteq W$ be an explicitly presented lattice. Set

$$\Delta_W = I - P_W.$$

The inclusion $\Lambda_B \subseteq W$ does not identify the discrete lattice with the real subspace.

Proposition 3.2 (Discrete CVP through harmonic projection)**.** *For $t \in \mathbb{R}^m$, put $t_W = P_W t$. Then:*

(i) t_W and every $\lambda \in \Lambda_B$ are Δ_W-harmonic;

(ii) for every $\lambda \in \Lambda_B$,

$$\|t - \lambda\|^2 = \|t - t_W\|^2 + \|t_W - \lambda\|^2;$$

(iii) a lattice vector λ is closest to t in Λ_B if and only if it is closest to t_W in Λ_B.

Proof. Since $t_W \in W$, every $\lambda \in \Lambda_B$ also lies in W, and $\ker(\Delta_W) = W$, the harmonicity claims follow. For the distance identity, write

$$t - \lambda = (t - t_W) + (t_W - \lambda).$$

Orthogonal projection gives $t - t_W \in W^\perp$, while $t_W - \lambda \in W$. The two summands are orthogonal, so the Pythagorean identity gives (ii). The term $\|t - t_W\|^2$ is independent of λ; consequently, for any $\lambda, \mu \in \Lambda_B$,

$$\|t - \lambda\|^2 \leq \|t - \mu\|^2 \quad \Longleftrightarrow \quad \|t_W - \lambda\|^2 \leq \|t_W - \mu\|^2.$$

This is exactly (iii). □

The continuous projection $t_W = P_W t$ is the closest harmonic vector to t, but it need not lie in Λ_B. Thus projection transports the target to the harmonic subspace while preserving the discrete minimizers; it does not perform the remaining lattice search. The proposition also presumes that W and the inclusion $\Lambda_B \subseteq W$ have been supplied. It does not discover the operative hidden subspace from public class data.

3.4 Oracle Insufficiency

The oracle-insufficiency direction uses the typed realization from Definition 2.12. A public Hodge solver may call an explicit-lattice CVP oracle and receives the actual public observation (α, b), but it receives no hidden representative or operative Laplacian. Correctness is required only for representatives admitted by the realization, rather than for every matrix of the ambient size.

Definition 3.3 (Public-data Hodge solver contract)**.** *Let*

$$S : \mathcal{O}_{\mathrm{CVP}} \times (P \times \mathbb{R}^n) \longrightarrow \mathbb{R}^n,$$

where $\mathcal{O}_{\mathrm{CVP}}$ is the class of explicit-lattice CVP oracles. For a realization $x \mapsto (M_x, \Delta_x)$ and $O \in \mathcal{O}_{\mathrm{CVP}}$, define

$$\begin{aligned} \mathsf{H}_O(S) &\iff \forall x : A, \quad \forall b \in \mathbb{R}^n, \quad \Delta_x\big(S(O, (\pi(x), b))\big) = 0, \\ \mathsf{F}_O(S) &\iff \forall x : A, \quad \forall b \in \mathbb{R}^n, \quad \Delta_x b = 0 \Rightarrow S(O, (\pi(x), b)) = b. \end{aligned}$$

The public-data-only contract is

$$\exists S, \quad \forall O, \quad \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow \big(\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)\big).$$

Proposition 3.4 (Typed HCP ambiguity and CVP oracle insufficiency)**.** *Let $x, y \in A$ be representatives in a typed HCP semantic realization. Suppose that M_x and M_y are homotopy equivalent and that some $b \in \mathbb{R}^n$ satisfies*

$$\Delta_x b = 0, \qquad \Delta_y b \neq 0.$$

If at least one correct explicit-lattice CVP oracle exists, then

$$\neg\exists S, \quad \forall O, \quad \mathsf{CVP}_{\mathrm{corr}}(O) \Rightarrow \big(\mathsf{H}_O(S) \wedge \mathsf{F}_O(S)\big).$$

Hence a correct oracle for explicit known-lattice CVP does not by itself determine the operative Hodge–Laplacian kernel.

Proof. By Proposition 2.13, $\pi(x) = \pi(y)$. Suppose $\exists O, \mathsf{CVP}_{\mathrm{corr}}(O)$, and choose such an O. Suppose, for contradiction, that S satisfies the displayed public-data-only contract. Then

$$\mathsf{H}_O(S) \wedge \mathsf{F}_O(S).$$

Since $\Delta_x b = 0$, exactness at x gives

$$S(O, (\pi(x), b)) = b.$$

Soundness at y gives

$$\Delta_y S(O, (\pi(y), b)) = 0.$$

Replacing $\pi(y)$ by $\pi(x)$ and using the exactness equality yields $\Delta_y b = 0$, contradicting the separation hypothesis. □

The obstruction is kernel ambiguity: the same public observation may be compatible with multiple hidden operators whose harmonic kernels differ. A CVP oracle can solve a closest-vector problem after the lattice is presented. The bridge in Proposition 3.2 applies only after the operative harmonic subspace and the lattice inclusion have also been supplied; it does not determine which hidden kernel is compatible with the public data.

For every dimension at least two, consider the elementary matrix witness $L = 0$, $L' = \mathrm{diag}(1, 0, \ldots, 0)$, and $b = e_1$ under the constant projection to a one-point public type. This is a sanity example showing that the abstract ambiguity predicates are inhabited. It is not used as evidence that an arbitrary HCP realization admits the required semantic and Hodge extension witness. That witness remains an explicit hypothesis of Proposition 3.4; the following family constructs one for its stated semantic fragment.

3.5 A Constructed Integral Cellular Family

The preceding implication has a concrete instantiation in a stated homological fragment of the typed Helfer interface. Take one vertex, one edge and one face with incidence numbers $\partial_1 = 0$, $\partial_2 = 1$, and stalks $0, \mathbb{R}^2, \mathbb{R}$, respectively. This is the algebraic finite signed-incidence model of Section 2; no regular CW or simplicial realization of this displayed base is asserted. Two cellular sheaves in that model have

$$\delta_{0,0} = \delta_{0,1} = 0, \qquad \delta_{1,0}(x_0, x_1) = x_0, \qquad \delta_{1,1}(x_0, x_1) = x_1.$$

Their true degree-one Hodge operators are

$$\Delta_0 = \mathrm{diag}(1, 0), \qquad \Delta_1 = \mathrm{diag}(0, 1).$$

Coordinate exchange is a degreewise chain isometry. Both operators have positive spectral gap 1. Standard integral stalk lattices give harmonic lattices $\mathbb{Z}e_1$ and $\mathbb{Z}e_0$, each with an explicit full-span presentation.

Interpret these degree-zero-through-two complexes in the bounded-below real cochain Quillen model category, with the incidence-defined coboundaries displayed above. The closed tests are indexed by integers and ask whether cohomology in that degree is nonzero. The required cofibrancy and fibrancy

hold for these real complexes. The model's weak equivalences are quasi-isomorphisms, which preserve the tests and give the required closed-test invariance under the declared typed Helfer equivalences.

The coordinate-exchange chain isometry therefore makes the two actual homological theories equal, supplying the shared public class used below. This establishes invariance for the declared fragment; equality of its class values need not reconstruct a homotopy equivalence or classify arbitrary hidden presentations. Helfer's full first-order formula syntax, equality objects and general fibrational invariance theorem remain external background.

Proposition 3.5 (One observation, two nonzero integral harmonic targets)**.** *Let p be the common closed theory of these two presentations in the stated fragment, and let $b = e_0 + e_1$. There are structured Hodge samples with that same observation (p, b), harmonic messages e_1 and e_0, and zero exact components. Their operative targets are distinct and nonzero. For each presentation's explicit harmonic lattice, its message is also the unique closest lattice vector to b.*

Proof. In the first presentation use harmonic message e_1, exact opening zero and coexact opening 1, producing noise e_0. In the second use message e_0, exact opening zero and coexact opening 1, producing noise e_1. Both sums equal b, and both messages belong to their standard integral harmonic lattices. The actual Hodge projections recover those messages. Orthogonality gives the Pythagorean identity from Proposition 3.2; any other lattice point has strictly positive squared distance from the recovered message. Thus each message is the unique CVP answer in its presented lattice. □

For the same realization, the fixed coordinate e_1 is harmonic for Δ_0 and not for Δ_1. This constructs the separation witness used in Proposition 3.4. Equality of the public theory is derived from the actual semantic equivalence, rather than imposed by a one-point classifier. The matrix-level public projection is extended from the two admitted operators; no semantic realization is asserted for other matrices.

The displayed pair uses coefficient 1 in each presentation's own harmonic-lattice basis. Its distinct ambient targets therefore do not alone assert distinct encoded integer plaintexts. A second constructed pair makes the encoding explicit: take $b = e_0 + 2e_1$, harmonic messages $2e_1$ and e_0, and complementary coexact noise e_0 and $2e_1$. Their integer coefficients in the same declared lattice presentations are 2 and 1. Both messages are nonzero, both exact components remain zero, and the public observation is again shared. No function of that observation can return the correct coefficient

for both presentations. This is a finite exact-recovery obstruction for the specified integer encoding, not a distributional security experiment. Each message remains the unique CVP answer in its actual lattice.

This example fixes the mathematical observation to (p, b). Its equal zero exact components do not imply equality of complete storage records. A larger released view must be specified and shown to agree for both presentations before the same ambiguity argument applies. Transferring the target and CVP conclusions also requires agreement with the operative operator and classifier. No distributional anonymity or plaintext-security conclusion follows from this example.

Corollary 3.6 (Strict separation for hidden lattice recovery)**.** *Fix a correct CVP oracle for the presented rank-one lattices of the constructed family. When the operative lattice is supplied, the oracle recovers its unique closest vector, equal to the operative harmonic target. From the shared public class and cochain alone, no solver with that same fixed oracle can recover the operative target for both hidden members. Every larger compatible family retaining those two members inherits this exact-recovery obstruction.*

Proof. For each supplied lattice, oracle correctness and Proposition 3.5 identify its answer with the harmonic message. The two hidden members have the same public observation and distinct operative targets. A solver receives the same observation and the same oracle in either case, so its single output cannot equal both targets. The argument allows arbitrary fixed auxiliary information, including all candidate CVP answers, and imposes no running-time bound. Any compatible family containing the pair retains this contradiction. □

Solving the candidate CVP instances does not select the operative HCP answer.

4 Conclusion and Further Work

The hidden Hodge operator determines the recovery target. An explicitly presented harmonic lattice gives a precise discrete-CVP comparison, while the constructed common-input examples prevent uniform exact recovery from the specified public data. Exact-component observations and full projective observations expose different information, as the two observation results make precise.

4.1 Further Questions

The comparison suggests studying encoded HCP families with explicit distributions on hidden presentations and Hodge-structured noise. Their computational difficulty must be assessed within that specified model.

Discrete Morse reduction for cellular sheaf cohomology [CGN15] suggests a computational refinement. A chain-homotopy equivalence alone does not identify the original harmonic representative or preserve its distance. An application to the original recovery target needs explicit retained lifts and a metric comparison.

The constructed classifier proves the named homological-test fragment in the real cochain model. Extending it to the full first-order fibrational semantics, constructing richer families with a declared topological realization, and analyzing transported public observations are separate mathematical tasks. The selected results remain the exact Hodge/CVP comparisons and observation boundaries of this manuscript.

A Statement–Lean Parity Table

Table 2 records the correspondence for this paper's six Hodge/CVP groups and two observation-boundary groups. The independent entry's configuration, `lean/Palomar/PaperOne/comparator.json`, selects 21 manuscript-facing and 25 supporting theorem declarations. Every declaration appears once below. These are coverage counts, not a count of independent research contributions. Definition rows name canonical implementation objects, not additional selected theorem targets. Canonical consequences identify derived implementation lemmas outside that configured selection. Main-result and support rows are proved under their stated premises; background, assumption and contract rows identify the separate scope boundaries.

Write `E` for `Hcrypt.Palomar.PaperOne`, `CS` for `E.CellularSheaf`, `HB` for `E.HCPBridge`, `AF` for `E.AxisFamily`, and `PL` for `E.Plucker`. The abbreviated anchors expand to the exact configured declaration names.

Table 2: Selected manuscript–Lean parity.

Statement, status and scope	**Corresponding formal declarations or boundary**
Definitions: HCP input and target. Section 2. The public pair contains a classifier value and observed cochain. A private representative selects the operative kernel and hence the orthogonal-projection target.	Canonical definitions: `Hcrypt.HCPBridge.Observation` `Hcrypt.HCPBridge.HiddenHodgeWitness` `Hcrypt.HCP.operativeHarmonicComponent` `Hcrypt.HCPBridge.operativeTarget`
Definitions: explicit-lattice CVP. Section 3.1. An injective real basis presents the lattice through integer coefficients. An instance adds a target; a solution is a closest discrete lattice point.	Canonical definitions: `CVP.ExplicitLattice` `CVP.Instance` `CVP.Solution`
Definitions: semantic realization. Section 2.2. The public classifier factors through the specified closed theory. The operative Hodge operator is separate presentation-level data; equal classifier values need not identify it.	Canonical definitions: `Hcrypt.Axioms.PublicClassifier` `Hcrypt.Helfer.ClosedTheory` `Hcrypt.HCPBridge.Realization`
Definitions: Hodge sample and lattice. Sections 2–3. The sample combines an integral harmonic message with exact/coexact noise. A supplied harmonic-lattice presentation has full real span; it need not contain every intrinsic integral harmonic cochain.	Canonical definitions: `Hcrypt.Sheaf.CellularSheaf.HodgeEncrypte dCochain` `Hcrypt.Sheaf.CellularSheaf.HarmonicLatti cePresentation`

Continued on the next page.

Table 2: Selected manuscript–Lean parity (continued).

Statement, status and scope	Corresponding formal declarations or boundary
Main result: P1-S01. *Proved.* Proposition 2.7; private recovery in Sections 2–3. The complex equation and actual adjoint Hodge operator give componentwise unique harmonic, exact and coexact parts. Integer coefficient recovery uses the supplied harmonic-lattice presentation.	`E.Hodge_Decomposition_Theorem` `E.Hodge_Decomposition_unique` `CS.HodgeEncryptedCochain.harmonicOneProjection_observed` `CS.HarmonicLatticePresentation.decodeCoefficients_encrypt`
Main result: P1-S02. *Proved.* Proposition 3.1. A field-linear image of the displayed generator is realized as a kernel. The consequence concerns that image, not integer-lattice membership.	`E.exists_matrix_ker_eq_range` `E.Hodge_GeneratedSubspace_Reduction_Proof`
Main result: P1-S03. *Proved.* Proposition 3.2. The discrete lattice is explicitly supplied and lies in the projected subspace. The Pythagorean identity preserves exactly its closest-vector answers.	`E.squaredDistance_projection_decomposition` `E.solution_iff_projectedTarget`
Main result: P1-S04. *Proved.* Cellular specialization of Proposition 3.2. For the actual Hodge sample and supplied harmonic message lattice, the operative message is the unique closest lattice point.	`E.concreteSheafHCP_CVP_solution_iff_message` `E.concreteSheafHCP_CVP_existsUnique`

Continued on the next page.

Table 2: Selected manuscript–Lean parity (continued).

Statement, status and scope	Corresponding formal declarations or boundary
Main result: P1-S05. *Proved.* Propositions 3.4 and 3.5 and Corollary 3.6. Generic separation retains closed-test invariance and kernel separation; oracle corollaries also retain existence of a correct explicit-lattice CVP oracle. The constructed integral family has one public observation and distinct nonzero targets. Its supplied rank-one lattice is solvable by that same correct oracle; shared-class recovery is impossible, as is exact recovery in every compatible family retaining the separated pair.	`HB.operativeTargets_ne_of_kernelSeparation` `HB.cvpOracle_insufficient_of_kernelSeparation` `HB.cvpOracle_insufficient_for_operativeTarget` `AF.cellular_axis_ambiguity` `AF.cvp_oracle_insufficient` Canonical consequences: `Hcrypt.HCPBridge.no_solver_exactOnOperativeTarget_on_family` `Hcrypt.HCP.AxisFamily.cvp_known_member_hidden_member_separation`
Main result: P1-S06. *Proved.* Weighted example in Section 3.5. The common observation is $b = (1, 2)$, with private integer coefficients two and one. Each privately presented lattice has its own unique CVP answer.	`AF.weighted_axis_recovery_obstruction`
Main result: P1-B01. *Proved.* Proposition 2.8. An allowed nonzero coexact direction prevents the full Hodge residual from factoring through the exact component. The farness obstruction uses the unrestricted noise carrier and exact-component-only extraction.	`CS.HodgeEncryptedCochain.not_residual_norm_factors_exactComponent` `CS.HodgeEncryptedCochain.not_far_preserving_exactComponent`

Continued on the next page.

Table 2: Selected manuscript–Lean parity (continued).

Statement, status and scope	Corresponding formal declarations or boundary

Main result: P1-B02. *Proved.* Proposition 2.17. A genuine independent frame's full projective coordinates recover its subspace and fixed-metric projector. Operative-target recovery additionally identifies that subspace with the operative kernel. A digest is a different observation.	`PL.recoverSubspace_frame`
	`PL.recoverHarmonicProjector_frame`
	`PL.recover_eq_operativeHarmonicComponent`

Support: complexes and semantics. *Proved.* Sections 2–3. Actual coboundary composition, the constructed complex isomorphism and the corresponding three-term differential.	`CS.delta1_comp_delta0`
	`AF.cellular_complex_iso`
	`E.ThreeTermData.differential_comp`
	`AF.boundarySqZero`

Support: axis coordinates. *Proved.* Section 3.5. Coordinate vectors and the distinct complementary-axis choices in the examples.	`AF.unitCochain_apply`
	`AF.other_zero`
	`AF.other_one`
	`AF.other_ne`

Support: axis Hodge operators. *Proved.* Section 3.5. Both coboundaries, the adjoint, the degree-one Laplacian and its harmonicity criterion.	`AF.delta0_eq_zero`
	`AF.delta1_apply`
	`AF.delta1_adjoint_eq`
	`AF.degreeOneLaplacian_apply`
	`AF.harmonic_iff`

Support: integral lattice. *Proved.* Section 3.5. Restriction-compatible stalk integrality, harmonic basis vectors, encoding injectivity and membership in the chosen lattice.	`AF.standardStalkLattice_contains_iff`
	`AF.restriction_preserves`
	`AF.unitCochain_integral`
	`AF.other_unit_harmonic`
	`AF.harmonicBasis_injective`
	`AF.harmonicLatticeVector_eq`
	`AF.latticeVector_mem`

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Table 2: Selected manuscript–Lean parity (continued).

Statement, status and scope	**Corresponding formal declarations or boundary**
Support: projective quotient. *Proved.* Section 2.2.1. Scalar-orbit equivalence laws, wedge-annihilator recovery and invariance under a nonzero scalar multiple.	`PL.ProjectivelyEquivalent.refl` `PL.ProjectivelyEquivalent.symm` `PL.ProjectivelyEquivalent.trans` `PL.mem_wedgeAnnihilator_iff` `PL.wedgeAnnihilator_eq_of_projectivelyEquivalent`
External theorem and semantic premise. Sections 2.2 and 3.5; distinct from the proved homological fragment.	Helfer's general fibrational/path-invariance theorem is cited background. Applying the generic bridge requires the declared invariant closed tests; full first-order invariance and a complete classifier for arbitrary hidden presentations are not proved by this entry.
Assumption: computational hardness. Section 3.1; distinct from the proved exact comparisons.	Known-lattice CVP hardness is an external computational interface. The selected algebraic comparisons and non-identifiability examples do not establish an asymptotic encoded-instance hardness reduction or a sampled encryption-security theorem.
Contract: protocol application. Section 2.2; distinct from the observation theorems.	A larger released view requires its own equality or factorization proof. Artifact binding, encoding, visibility and source-to-code metric comparisons are additional application conditions. No general full-HCP distance comparison follows from an exact-component extraction.

B Formal Objects and Proof Boundaries

Independent statements and canonical proofs. `Palomar.PaperOne.Challenge` gives ordinary transparent definitions using Mathlib-only imports. Its finite-product helpers `Hcrypt.Helfer.contextObject` and `Hcrypt.Helfer.contextMap` retain their canonical names. `Palomar.PaperOne.Solution` identifies the independent presentations with the canonical proof objects without importing Challenge. The 23 deliberate Challenge theorem placeholders are comparison targets; Solution has no proof or definition holes.

Source objects and mathematical scope. The finite cellular Hodge operators and integral message presentation are in the `Sheaf` modules. `CVP.lean`, `HCP.lean` and `Hardness.lean` supply the lattice and operative-target comparisons. `Sheaf/HCPAxisFamily.lean`, `HCP/AxisFamily.lean` and `Helfer/CochainModel.lean` construct the actual integral examples and homological-test interpretation. They use the algebraic finite signed-incidence model; no regular CW or simplicial realization of its base is asserted. Integer coefficient recovery is classical and noncomputable. The proved semantic fragment does not supply the general first-order theorem or arbitrary-presentation classifier.

The full-point reconstruction is owned by `Proximity/Concrete/GrassmannianPlucker/Reconstruction.lean`; `TCB/PluckerObservation.lean` identifies the operative target under explicit kernel agreement. The raw projective quotient contains the zero orbit. Recovery uses genuine independent frames, full exact-real coordinates and the fixed ambient metric; it supplies neither digest inversion nor an efficient finite-byte decoder.

Validation applies to exact snapshots. The permitted proof axioms are `propext`, `Quot.sound` and `Classical.choice`; the selected dependency audit reports only these axioms or subsets. The earlier public snapshot `66e1ad2` passed protected preflight. That result does not certify this revised manuscript; its eventual release requires its own exact-commit comparison and report. Mechanical checking is separate from editorial review, novelty assessment and registry registration. The release manifest records the proof/import closure and hashes; the parity table identifies the selected results, rather than every declaration in those supporting modules.

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